

HW4

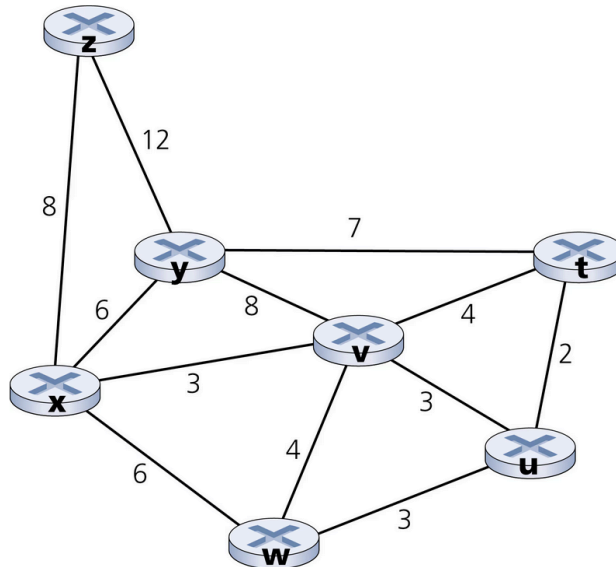
1. Is it necessary that every autonomous system use the same intra-AS routing algorithm? Why or why not? (R1)

No. autonomous systems are independent so they can run whatever internal routing protocol they want. They only need BGP for external routing.

2. True or false: When a BGP router receives an advertised path from its neighbor, it must add its own identity to the received path and then send that new path on to all of its neighbors. Explain. (R13)

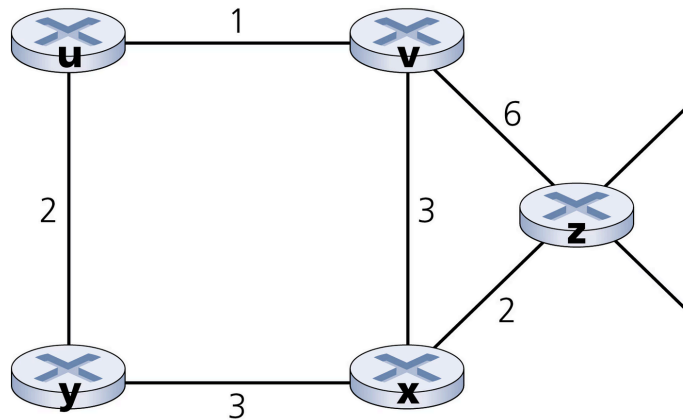
False. BGP uses import and export policies to filter paths based on business rules.

3. Compute the shortest path from *v* to all other nodes using Dijkstra's shortest path algorithm. Show a table similar to the example we did in class. (P4)



Step	N'	D(t), p(t)	D(u), p(u)
0	v	4, v	3, v
1	v,u	4, v	
2	v,u,x	4, v	
3	v,u,x,t		
4	v,u,x,t,w		
5	v,u,x,t,w,y		
6	v,u,x,t,w,y,z		

4. Consider the network shown below, and assume that each node initially knows the cost to each of its neighbors. Consider the distance-vector algorithm and show the distance table entries at each node. You do not need to show intermediate steps, just the table entries at each node. (P5)



	v	x	y	z
u	(1,v)	(4,v)	(2,y)	(6,v)
v	(1,u)	(3,x)	(3,u)	(5,x)
x	(4,v)	(3,v)	(3,y)	(2,z)
y	(2,u)	(3,u)	(3,x)	(5,x)
z	(6,x)	(5,x)	(2,x)	(5,x)

5. Describe how loops in paths can be detected in BGP. (P12)

by checking the AS-PATH attribute. If a router sees its own ASN in the path it drops it to prevent loops.

6. Will a BGP router always choose the loop-free route with the shortest ASpath length? Justify your answer. (P13)

No. local preference is evaluated first to enforce business policies. The shortest AS-PATH is just a tiebreaker.

7. Consider the network shown below. Suppose eBGP and iBGP are used for the inter-AS routing protocol. (P14)

- Router 3c learns about prefix x from which routing protocol: IGP, eBGP, or iBGP?
eBGP
- Router 3a learns about x from which routing protocol?
iBGP
- Router 1c learns about x from which routing protocol?
eBGP
- Router 1d learns about x from which routing protocol?
iBGP

